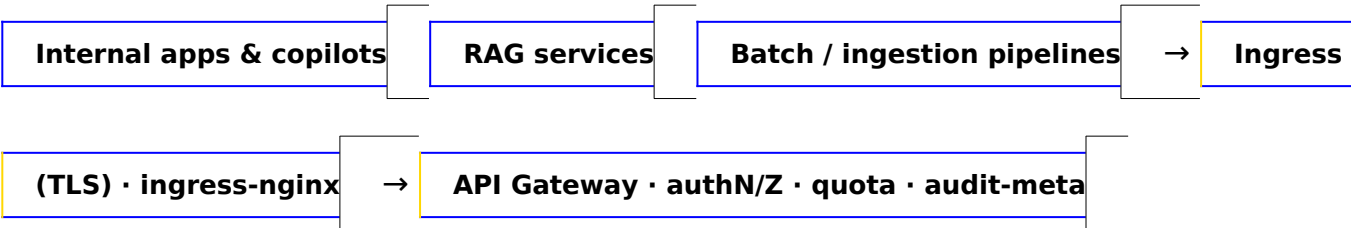


6-Blade Kubernetes AI Platform — Detailed Architecture

Supermicro SBE-820C-822 chassis · 6× SBI-8149P-C4N (CPU-only) running RKE2 · GLM-5.2 inference on a separate GPU plane

- 6 Blades (3 control-plane + 3 workers)
- 3 etcd members — HA quorum
- ≥20% Reserved RAM headroom / node
- 1→2 GPU servers (HA on roadmap)

1Consumers & Entry (North–South)



TLS terminates at ingress; the gateway authenticates every request, enforces quotas, and records metadata

only — never prompt/response bodies.



2Plane 1 — Physical Blade Allocation (RKE2 Kubernetes)

■ Supermicro SBE-820C-822 Chassis — dual redundant switch modules CPU-

only · ~1,000 cores · ~2 TB RAM aggregate (per-blade split = assumption,

confirm in discovery)

blade-01 10.20.0.11

Control plane · tainted NoSchedule

kube-apiserver · scheduler · controller-mgr

etcd member 1 · dedicated NVMe

ingress-nginx

Reserve: OS + kubelet + ≥20% RAM headroom

blade-02 10.20.0.12

Control plane · tainted NoSchedule

kube-apiserver · scheduler · controller-mgr

etcd member 2 · dedicated NVMe **Vault**

· cert-manager · Harbor

Reserve: OS + kubelet + ≥20% RAM headroom

blade-03 10.20.0.13

Control plane · tainted NoSchedule

kube-apiserver · scheduler · controller-mgr

etcd member 3 · dedicated NVMe

Prometheus · Loki · Grafana · Argo CD

Reserve: OS + kubelet + obs + $\geq 20\%$

blade-04 10.20.0.14

Worker · AI apps

API Gateway (Envoy/APISIX) **LiteLLM model router** → GPU + CPU models

Orchestrator / RAG / agents Guardrails / PII redaction

Guaranteed QoS · anti-affinity · $\geq 20\%$ headroom

blade-05 10.20.0.15

Worker · data + CPU models

Qdrant vector DB (persistent NVMe) **Redis** cache / rate-limit store

Embeddings (text-embeddings-inference, BGE-M3) NUMA-pinned · CPU

Guaranteed QoS · $\geq 20\%$ headroom

blade-06 10.20.0.16

Worker · CPU models + burst

Rerankers (cross-encoder, ONNX/TEI) **Embeddings replica Ingestion**

workers + queue (NATS)

Batch / burst capacity

Guaranteed QoS · $\geq 20\%$ headroom

Control plane / etcd Worker (microservices) etcd (dedicated NVMe) Security Observability Data AI service

3Cluster-Wide Platform Layers (East–West)

Orchestration & Networking

•RKE2 (Kubernetes) · containerd

•Cilium eBPF CNI · default-deny NetworkPolicy

•CoreDNS · kube-vip (API VIP 10.20.0.10)

•KEDA + HPA (scale on queue/latency, not CPU%)

Security & Supply Chain

•Vault (secrets) · cert-manager (internal CA)

•Kyverno admission — signed + digest-pinned images

•Harbor registry · Trivy scan · cosign/Notation

•PSA restricted · mTLS east-west · audit log

GitOps · MLOps · Observability

•Argo CD (app-of-apps, pinned tags)

•Model registry (metadata) + MinIO (weights)

•Prometheus · Loki (redacted) · Grafana · OTel

•Velero backups → MinIO

4Network Segmentation & Storage

VLAN segmentation

Mgmt / IPMI — BMC, provisioningisolated

K8s node / API — kubelet, etcd peerlow-latency

Pod / CNI — Cilium overlaydefault-deny

Storage — MinIO / NVMe, jumbo MTU 9000VLAN

Inference VLAN — blades ↔ GPU (mTLS only)firewalled

Ingress — consumers → gatewayTLS

Storage design (no default Ceph)

•**etcd** → dedicated local NVMe per CP (p99 fsync < 10 ms)

•**Vector DB / Redis** → local NVMe persistent volumes

•**MinIO** (on blades) → object store: model master copy (signed + checksummed), documents, backups

•**Model serving copy** → pulled & verified to GPU host local NVMe

•**Observability** → local NVMe + MinIO long-term tier

5Plane Boundary → Plane 2 GPU Inference (Separate Server)

LiteLLM router (blade-04) → mTLS + bearer token · OpenAI-compatible ·

inference VLAN → **ExternalName / Endpoints → gpu-infer 10.30.0.20:8000**

Cilium egress policy allows only router pods to reach the GPU host. Only a URL + credentials cross the

boundary — the GPU backend is swappable without touching any microservice.



One GPU server = single point of failure for the large model; a 2nd backend (router failover) delivers

inference HA and ~2× throughput — GPU expansion roadmap.

Evidence tags — [DOC] documented · [EST] estimate, benchmark to confirm · [ASSUMPTION] pending Phase-0 hardware discovery. All data & prompts remain on-premises. Provisional until discovery/benchmark replace assumptions with measured facts. · Enterprise Architecture · July 2026